

Program : Diploma in Chemical Engineering	
Course Code : 3076	Course Title: Chemical Equipment Drawing
Semester : 3	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L: 0 T: 0 P: 3)	Periods per semester: 45

Course Objectives:

- To enable the students to interpret the chemical engineering drawings commonly used in Industries.

Course Pre-requisites:

Topic	Course code	Course name	Semester
Engineering Graphics		Engineering Graphics	1
Computing fundamentals		Introduction to IT systems	1
Fundamentals of AutoCAD		CAD Lab	2

Course Outcomes :

On completion of the course student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Sketch fastening devices and joints.	15	Applying
CO2	Sketch vessels and vessel support.	15	Understanding
CO3	Sketch heat transfer equipment.	6	Applying
CO4	Sketch mass transfer equipment.	9	Understanding

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1		3		3			
CO2			2	3			
CO3			3	3			
CO4			2	3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcome	Description	Duration (Hours)	Cognitive Level
CO1	Sketch fastening devices and joints.		
M1.01	Sketch square nuts, hexagonal nuts and bolts using ASME code B 18.2.1	3	Applying
M1.02	Sketch riveted joints.	3	Understanding
M1.03	Sketch Flanged pipe joint using ASME code B16.5	3	Applying
M1.04	Sketch Socket and Spigot joint.	3	Understanding
M1.05	Sketch muff coupling, flanged coupling.	3	Understanding
CO2	Sketch vessels and vessel support.		
M2.01	Sketch vessel supports.	3	Understanding
M2.02	Sketch Cone roof tank	3	Understanding
M2.03	Sketch Internal floating roof tank and External floating roof tank.	3	Understanding
M2.04	Sketch pressure vessels.	3	Understanding
M2.05	Sketch Storage bullets.	3	Understanding
	Series Test – I		
CO3	Sketch heat transfer equipment.		
M3.01	Sketch 1-2 pass shell and tube heat exchanger for a given specification.	3	Applying
M3.02	Sketch vertical short tube evaporator	3	Understanding

CO4	Sketch mass transfer equipment.		
M4.01	Sketch a tray tower for distillation.	3	Understanding
M4.02	Sketch a packed tower for absorption.	3	Understanding
M4.03	Open ended experiments	3	Applying
	Series Test – II		

Text / Reference

T/R	Book Title/Author
T1	Process Heat Transfer & Chemical Equipment Design; D.C. Sikdar.
R1	Joshi's Process Equipment Design; V. V.Mahajani and S. B. Umarjii.
R2	Chemical Engineering Design", Coulson and Richardson's Chemical Engineering Series, Volume-6; R. K. Sinnott.
R3	Chemical Engineers' Handbook; R. H. Perry,
R4	Introduction to Chemical Equipment Design Mechanical Aspects; B.C. Bhattacharyya.

Online Resources

Sl.No	Website Link
1	https://nptel.ac.in/courses/103/107/103107143/